

Mykola Zhuk

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📁 Experience

Software Developer - Client MVP

Jun 2026 – Present | Remote

Independent project for a startup founder

- **Independently built** a Python pipeline for **20 corporate-event categories**, processing **2,936 articles** into **1,332 records**, with **99.7% passing schema validation**; implemented ingestion, extraction, evaluation, and graph-ready exports.
- **Deployed and demonstrated the MVP** in an Azure development environment using Container Apps Jobs, Service Bus, Blob Storage, and PostgreSQL-backed idempotency; added automated tests, audit records, and hard inference-budget limits.
- Selected Mistral Small 3.2 24B through project-specific model evaluation; **reduced token usage by 17.9%** through cached classification, candidate filtering, and selective model calls.
- Improved event-candidate **precision from 45.0% to 58.6%** at approximately **95% recall** on a **300-article holdout**, comparing candidate-selection strategies through controlled experiments.

Backend Engineer - Volunteer

Oct 2025 – Mar 2026 | Remote

Tacmedbat

- Built NestJS/MongoDB APIs and a **custom admin interface** for editable Ukrainian/English website content; aggregated 8 collections into one public endpoint serving **10+ frontend blocks**.
- Delivered **15+ features and fixes** across content and image handling, role-based access, payments, and SMTP contact-form delivery; optimized the public-content retrieval path with sorting, archive handling, and hashing

Software Analyst

Jun 2024 – Aug 2024 | Toronto

Accenture

- Developed Python/OpenAI API prototypes for retail Q&A and semantic search, supported by data pipelines ingesting and enriching **50,000+ product records**.
- Delivered backlog features through Agile sprints, code reviews, and stakeholder demos with a cross-functional team.

📁 Projects

Full-Stack Engineer

Oct 2024 – Dec 2024 | Toronto

Scriptorium-CSC309 Project

- Implemented template, viewing, forking, and code-execution **endpoints and their frontend flows** in a team-built Next.js/React application using Prisma and PostgreSQL.
- Built Docker language environments with **execution-time and memory limits for 10+ languages**; deployed the application on AWS EC2.
- Added **GitHub and Google OAuth 2.0 sign-in**; tested API endpoints with Postman and verified responsive layouts.
- **Live Demo (AWS)**

Backend Engineer

Oct 2024 – Dec 2024 | Toronto

ProjectOllon (Industry-Partnered MVP) - CSC301 Project

- Designed a Prisma schema for **100+ files and 2,000+ code components**, **reducing full-scan time by 80%** in project evaluation.
- Integrated static analysis with OpenAI/Cohere models to flag duplication and complexity; helped deliver the code-review MVP to industry partner Ollon.

🧠 Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, Bash

Web & Data: React, Next.js, Node.js, NestJS, PostgreSQL, MongoDB, Prisma

Cloud & testing: Azure, AWS EC2, Docker, GitHub Actions, pytest, Jest, Grafana Loki, Postman, Git, Linux

🎓 Education

Honours Bachelor of Science, Mathematics

Sep 2020 – Jun 2025 | Toronto

University of Toronto

Relevant Courses: Software Design, Machine Learning, Data Structures and Algorithms, Web Programming